FINITENESS AND INTEGRALITY DESCEND ALONG A SEPARATED MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof handles both assertions simultaneously by writing 'finite (resp. integral)' at each step; the two parts share the whole argument.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Stability of finiteness and integrality under base change. Lemma lemma-base-change-integral, used at proof lines 9538-9540. Makes X x_Z Y -> Y finite (resp. integral).
- A section of a separated morphism is a closed immersion. Lemma lemma-section-immersion, used at proof lines 9541-9543. Makes the graph X -> X x_Z Y a closed immersion.
- A closed immersion is finite and integral. Lemma lemma-closed-immersion-finite, used at proof lines 9544-9545. Gives the second factor.
- Stability of finiteness and integrality under composition. Lemma lemma-composition-integral, used at proof lines 9546-9548. Concludes for f.

Source reference. Original lemma 03ZS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5231–5231. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 081Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10233–10233. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03ZR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5230–5230. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4685–4685. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of algebraic spaces over S.*

(1) *If $g \circ f$ is finite and g separated then f is finite.*
(2) *If $g \circ f$ is integral and g separated then f is integral.*

Proof. Assume $g \circ f$ is finite (resp. integral) and g separated. The base change $X \times_Z Y \to Y$ is finite (resp. integral) by Lemma 03ZS. The morphism $X \to X \times_Z Y$ is a closed immersion as $Y \to Z$ is separated, see Lemma 03KP. A closed immersion is finite (resp. integral), see Lemma 081Y. The composition of finite (resp. integral) morphisms is finite (resp. integral), see Lemma 03ZR. Thus we win. □